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An Early Cretaceous paleomagnetic pole from Ponta Grossa dikes (Brazil): Implications for the South American Mesozoic apparent polar wander path

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Abstract. Paleomagnetic results from the Early Cretaceous Ponta Grossa dike swarm are reported. These dikes crosscut sediments and basement rocks associated with the Ponta Grossa Arch, a tectonic feature in the eastern border of the Paleozoic-Mesozoic Paraná Basin. These dikes are vertical or subvertical and trend mainly NW and NE. Samples from 127 dikes were submitted to both alternating field and thermal demagnetizations; 121 of them yielded characteristic directions interpreted as original thermal remanent magnetizations. Both normal and reversed polarities as well as intermediate directions (6 dikes) are recorded. The paleomagnetic pole is located at 30.3°E, 82.4°S ($N = 115$; $\alpha_{95} = 2.0°$; $k = 43.8$). This pole along with the available paleomagnetic poles for the Paraná Basin volcanics (Serra Geral Formation), recalculated in this paper in the light of reported $^{40}Ar/^{39}Ar$ ages describe an apparent polar wander path segment corresponding to a clockwise rotation (~6.9°) of the South American plate. This movement is compatible with the initial phase of the South Atlantic opening with rifting progressing from south to north. However, the calculated plate velocity is higher than the predicted velocities reported so far based on oceanic floor magnetic anomalies. The new data reported in this paper allowed the calculation of a mean Early Cretaceous pole (60.3°E, 83.8°S; $N = 7$, $\alpha_{95} = 2.9°$) which can be discriminated from the Middle-Late Jurassic (191.3°E, 86.2°S; $N = 3$, $\alpha_{95} = 7.8°$) and Late Cretaceous (346.5°E, 84.1°S; $N = 5$, $\alpha_{95} = 4.9°$) mean poles for South America, indicating a more complex movement of the continent than the simple east-west drift generally proposed.

Introduction

It is currently believed that the South American plate underwent only east-west movements from Late Jurassic [e.g., *Beck*, 1988; *Castillo et al.*, 1991] through Late Cretaceous as the South Atlantic rifting propagated from high to lower latitudes. This belief was supported by a rather poor paleomagnetic database with scattered poles for the stable part of the continent. Until recently, most of the poles were based on few independent sites, and ages are normally poorly defined. Even when radiometric data are available, they are often unhelpful since the K-Ar determinations normally place the pole age within a broad interval (up to >20 Ma; see Table 2) of uncertainty. For these reasons many authors prefer to use a single mean reference pole for the whole Cretaceous and part of the Jurassic instead of defining an apparent polar wander path (APWP) from an age sequence of poles.

The Lower Cretaceous Paraná magmatic province in southeastern Brazil (the Serra Geral Formation) offered the possibility for improving the Early Mesozoic data set [*Ernesto and Pacca*, 1988; *Ernesto et al.*, 1990] allowing the calculation of 20 poles from extrusive rocks. Although these poles satisfy *Beck's* [1988] selection criteria, the APWP segment they draw was only indicative due to the paucity of a better age constraint. Recently, a large number of $^{40}Ar/^{39}Ar$ determinations for the Paraná rocks [*Renne et al.*, 1992a, b] became available, most of them obtained from the same samples used in the paleomagnetic work. Additional data [*Turner et al.*, 1994] have been published, mainly from subsurface samples. All these data show that only a few million years (~133-130 Ma) were required for the emplacement of the major part of the igneous rocks in the Paraná Basin.

The final stage of the huge Paraná magmatism corresponds to the Early Cretaceous Ponta Grossa intrusive activity [*Piccirillo et al.*, 1988, 1990; *Ussami et al.*, 1991], although available $^{40}Ar/^{39}Ar$ data ranging from ~135 to 127 Ma [*Renne et al.*, 1993; *Turner et al.*, 1994] do not show this time relationship clearly. Previous paleomagnetic works on the Ponta Grossa dikes [*Ernesto and Pacca*, 1988; *Raposo and Ernesto*, 1989; *Ernesto et al.*, 1990] also support the above statement. However, the paleomagnetic poles presented in those papers (including the pioneer work of *Creer* [1962]) refer to few sites relative to the overall dike swarm and indicated a large, and probably unrealistic, apparent polar shift.

The purpose of this paper is to present a new paleomagnetic data set for the Ponta Grossa (PG) dike swarm which supersedes all previous work and to reevaluate the paleomagnetic poles derived from the Paraná Basin rocks in view of the available $^{40}Ar/^{39}Ar$ dating in order to propose a better defined APWP segment for the Lower Cretaceous of South America.

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Geological Setting

The Ponta Grossa Arch is a large (~134,000 km²) tectonic feature on the eastern border of the Paleozoic-Mesozoic Paraná Basin (Figure 1), with north and south limits corresponding to the Guapiara and Rio Piquiri lineaments, respectively. This arch was most active during the Jurassic and Lower Cretaceous, when its structures were enhanced and the arch took its present configuration [*Almeida*, 1986]. Associated with this structure is one of the largest Phanerozoic mafic dike swarms in Brazil: the Ponta Grossa dike swarm. Although maps based on aeromagnetic data (Figure 1) show a high concentration of dikes in the central area of the arch (between Rio Alonzo and São Jerônimo-Curiúva lineaments), dike outcrops are mainly concentrated in the regions of Fartura, Sapopema, and Telêmaco Borba, where they intrude the Paleozoic sedimentary rocks of the Paraná Basin, and in the regions of Curitiba and Guapiara, where they intrude the Precambrian rocks of the crystalline basement. Results from integrated ground/airborne magnetometry [*Ussami et al.*, 1991] demonstrate that the magnetic anomalies currently associated with dikes are actually related to structural features in the crystalline basement. Dike thicknesses vary from tens to hundreds of meters and are often tens of kilometers in length. They are fine to medium grained, and no difference in grain size is evident for dikes cutting Paleozoic or Precambrian terrains. The coarsest-grained diabase was found intruding sediments in the western part of the arch, near Sapopema (Figure 1). The Ponta Grossa dikes trend mainly NW and dip vertically or near vertically; however, NE trending dikes are also found.

The Ponta Grossa dikes as well as the Mesozoic volcanic rocks of the Paraná Basin (Paraná magmatic province) are related to the breakup of western Gondwanaland, having been emplaced during the initial stages of the South Atlantic rifting [*Almeida*, 1986; *Piccirillo et al.*, 1988]. Chemically, they are mainly andesitic basalt tholeiites and rare rhyolite of the Chapecó type [*Piccirillo et al.*, 1990; *Benini*, 1992]. These authors distinguished two main groups for the basaltic dikes: a dominant, high-TiO_2 (> wt.2%) group and a subordinate, low-TiO_2 (< wt.2%) group, characterized for similar MgO content, by high and low incompatible-element contents, respectively. These rocks are intimately related to the flood volcanics of the northern Paraná magmatic region (north of Rio Piquiri lineament; Figure 1) as defined by *Bellieni et al.* [1984]. Albeit rarely, PG dikes are seen cross-cutting the Paraná flows, mainly in Fartura region.

Sampling Procedures and Analytical Work

Samples were collected generally along road cuts. Most samples are hand blocks, but also some cylinders were taken with the aid of a gasoline-powered rock drill. Sample orientation was performed by both magnetic and solar compasses whenever possible. Normally three samples (or six to nine cylinders) were collected from each dike (site) distributed through the exposed width. In favorable outcrops a more detailed sampling was done including both margins as well as the center. About 400 samples were taken from 127 dikes widely distributed in the swarm.

For paleomagnetic measurements samples were cut into specimens of 2.5-cm diameter and 2.2-cm height. At least three specimens from each site (dike) were analyzed. Remanent magnetization was measured using a Molspin Ltd. spinner magnetometer. Samples were submitted to both alternating field (af) and thermal demagnetization. For af cleaning both tumbling Molspin Ltd. and static (Sapphire Instruments) demagnetizers were used. In pilot specimens af cleaning was performed from fields of 5 mT up to 80 or 100 mT in steps of 5 mT. In general, samples were thermally demagnetized from 150° up to 580°C in steps of 50°C up to 500°C and in steps of 20° or 30°C up to 580°C, if necessary. An af field of 10 mT or higher was applied before the thermal cleaning in order to remove the softer viscous remanent magnetizations. Possible changes in susceptibility values were monitored using a Sapphire Instruments SI-1 susceptibility meter.

The demagnetization data have been analyzed by means of Zijderveld plots [*Zijderveld*, 1967] and Wulff stereograms, as seen in Figure 2. The majority of the samples revealed high magnetic stability (Figures 2a to 2d) and no important low-stability magnetization was detected. For these samples, two or three other specimens were submitted to fields of 15 to 35 mT or temperatures of 200° to 450°C. A few samples showed low-stability component of magnetization (Figure 2e) which was easily eliminated at temperatures above 400°C. In these cases the companion specimens (two or three) were submitted to a more detailed procedure. Samples from only six dikes gave inconsistent results and were rejected.

Ore microscopy observations show that the magnetic minerals in the PG dikes are mainly phenocrysts and groundmass crystals of titanomagnetites oxidized at high temperatures as indicated by typical textures like trellis, composite, and sandwich [*Haggerty*, 1981] and by phenocrysts and microphenocrysts of ilmenite. Typical thermomagnetic curves (Figure 3) for samples of different rock types and remanences from the various regions of the Ponta Grossa Arch show high Curie temperatures ranging from 500° to 580°C also indicating that the main magnetic carriers are Ti-poor titanomagnetites.

Results

The mean magnetization direction of each dike (Table 1) was obtained by averaging directions from the corresponding specimens. As can be seen in Figure 4, PG dikes show normal and reversed polarity as well as some intermediate directions (sites 5, 16,17, 18, 19, and 118), probably related to polarity transitions or geomagnetic excursions, indicating that these dikes were magnetized at least in two polarity intervals. However, the majority of sites (79 of 115) were magnetized while the geomagnetic field had normal polarity.

The distribution of the magnetization directions corresponding to the NW and NE trending dikes (Figure 5) are similar, both including normal and reversed directions and suggesting that both fracture systems were fed contemporaneously. Also, there is no relationship between magnetization directions and the chemical groups

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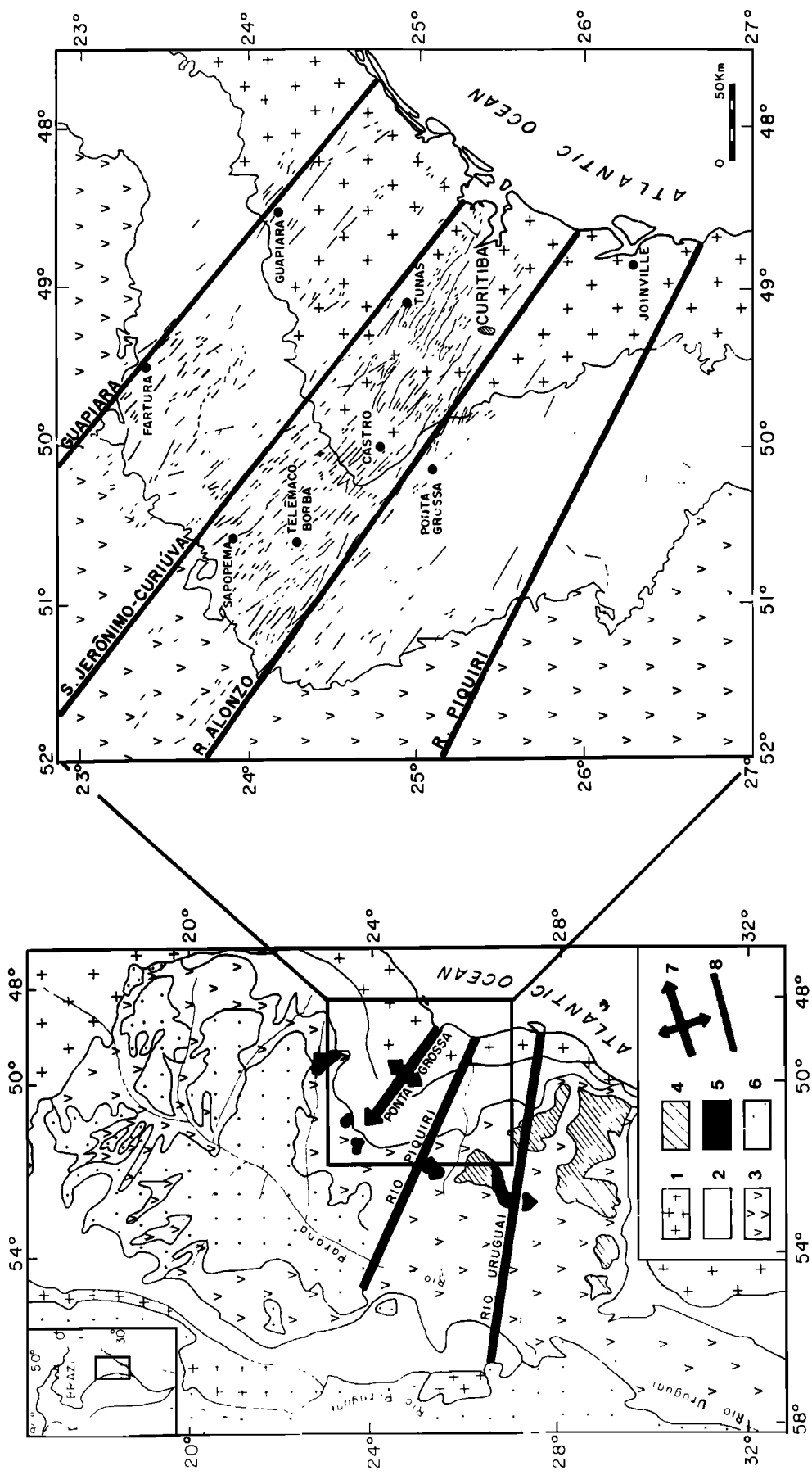


Figure 1. Generalized geological map of the Paraná Basin showing the Ponta Grossa Arch. Numbered features are (1) crystalline basement, (2) prevolcanic sediments (mainly Paleozoic), (3) basic to intermediate flood volcanics from the Serra Geral Formation. (4) acid lava flows (Palmas type), (5) acid lava flows (Chapecó type), (6) postvolcanic sediments (mainly upper Cretaceous), (7) arch-type structure, and (8) tectonic and/or magnetic lineaments.

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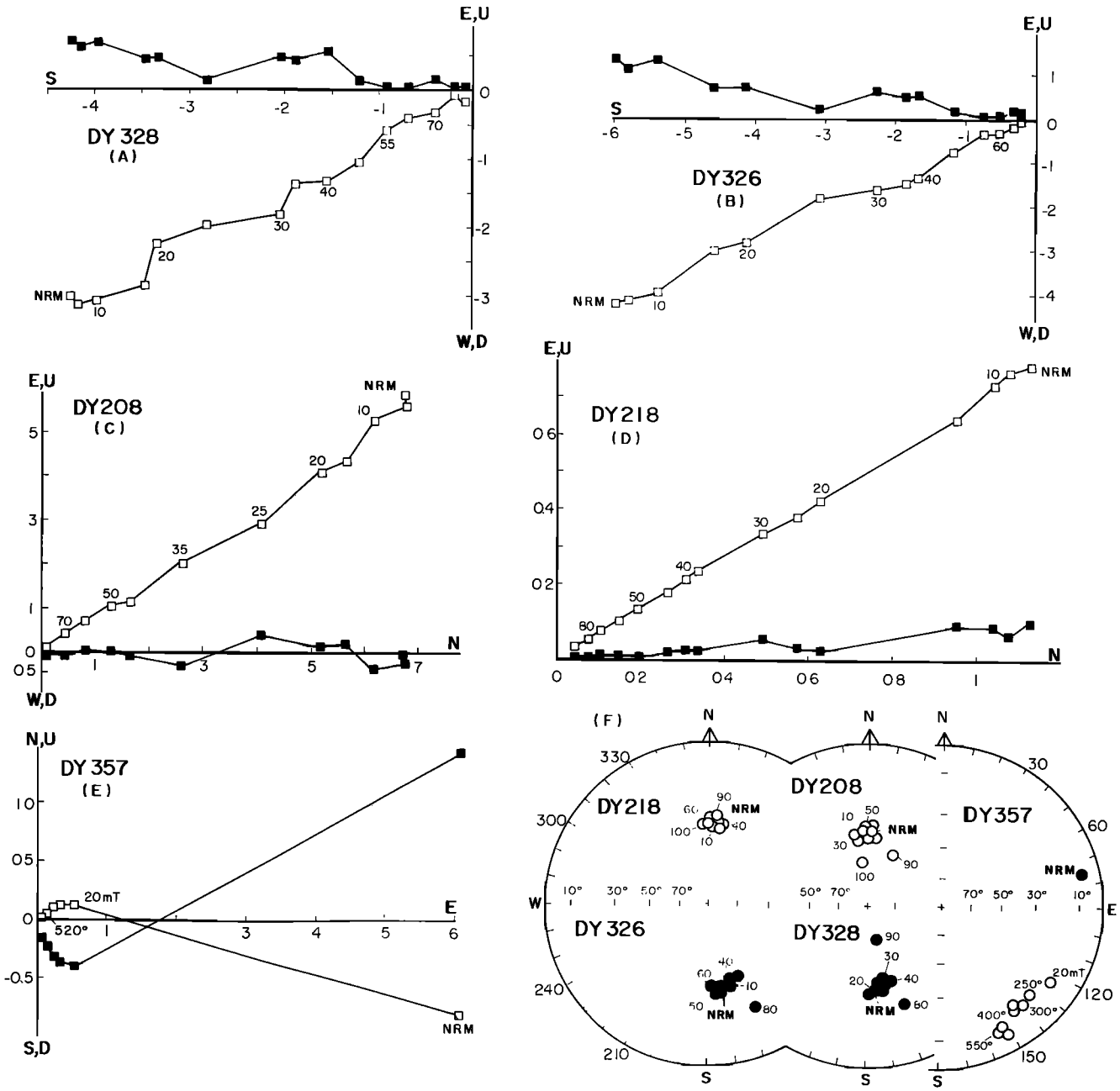


Figure 2. Zijderveld diagrams for the analyzed samples. (a) DY328, (b) DY326, (c) DY208, (d) DY218, and (e) DY357, where solid and open symbols are horizontal and vertical projections, respectively; magnetization intensities are in SI units; N is north, S is south, E is east, W is west, U is up, and D is down; and (f) magnetic behavior during the demagnetization process; solid and open symbols indicate positive and negative inclinations, respectively. Thermal demagnetization is in degrees Celsius, and alternating field is in milliteslas.

defined on the basis of TiO_2 content [*Piccirillo et al.*, 1990; *Benini*, 1992]. These relations were investigated in detail by *Raposo* [1992].

The degree of magnetic susceptibility anisotropy in the PG dikes [*Raposo and Ernesto*, 1995] is generally low (mean degree = 2.6%) showing that no deviations in the magnetization directions are expected. Actually, the mean magnetization direction for the PG dikes is nearly N-S (declination = 351.7, inclination = -42.7; $N = 115$; $\alpha_{95} = 2.0°$; $k = 42$) and does not lie close to the dike planes.

The paleomagnetic pole (coded PG in Table 2) calculated for the PG dikes is located at 30.3°E, 82.4°S ($N = 115$; $\alpha_{95} = 2.0°$; $k = 43.8$) and was computed as the mean of the virtual geomagnetic poles (VGPs) giving unit weight to each site, excluding those with intermediate polarity (six dikes).

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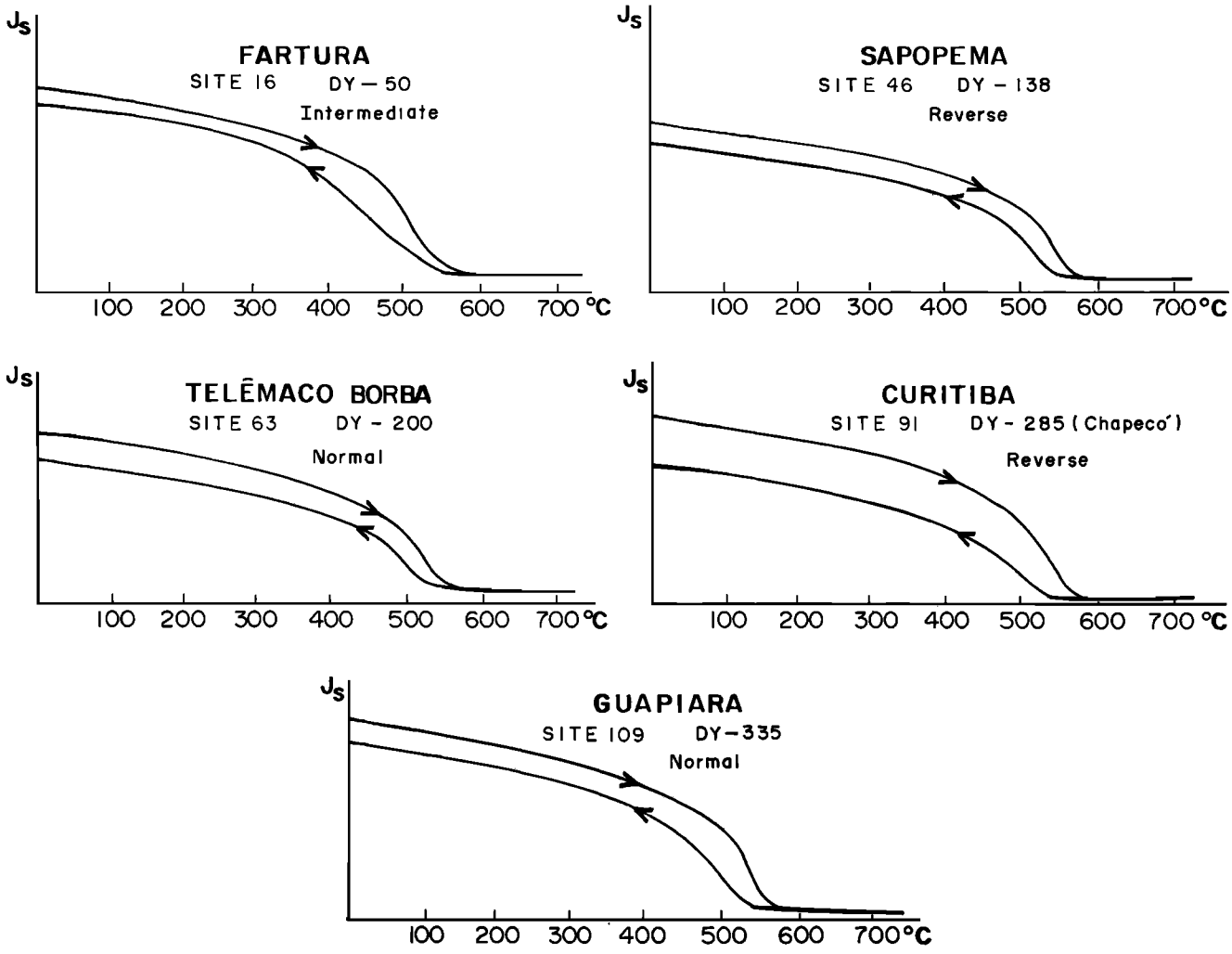


Figure 3. Examples of the thermomagnetic curves obtained for the Ponta Grossa (PG) dikes. The heating rate was 50 C min^{-1} in all cases.

Secular Variation

In order to evaluate whether geomagnetic secular variation has been averaged out from the VGPs, the angular dispersion S was compared with the expected angular dispersion [*McFadden and McElhinny*, 1984] at the corresponding paleolatitudes. The VGP angular dispersion for the PG dikes is $S = 12.2$ ($S = 81°/k^{1/2}$ [*McFadden*, 1980a]) with 95% confidence limits at 13.4° and 11.2°, according to *Cox* [1969]. As can be seen in Figure 6, the calculated VGP angular dispersion is in a relatively good agreement with curves for 0-5 Ma and 45-110 Ma age ranges but is lower than the predicted value in the 110-195 Ma curve, which is more appropriate for the age of the PG dikes. However, it must be considered that the secular variation model curve for 110-195 Ma was determined from only three data sets each with large 95% confidence limits. Thus the secular variation model for the 110-195 Ma interval is not well established.

The shape of the distributions of magnetization directions and VGPs (Figure 7) were tested by comparing them to a Fisherian distribution (Figure 8). For the magnetization directions (Figure 8a) the distribution of the angular distances from the mean is similar to the expected distribution only for angular distances greater than 5°. The χ^2 test of *McFadden* [1980b] indicated good azimuthal symmetry but no radial coherence at the 5% significance level. On the other hand, for the VGPs, both azimuthal (Figure 7b) and radial distribution (Figure 8b) show good probability ($p > 30\%$) at the 5% significance level. Therefore the VGP distribution can be considered approximately Fisherian, although there is a larger cluster of data within 5°-10° from the mean than would be expected from a Fisherian distribution (Figure 8b). According to *Baag and Helsley* [1974], if the distribution of the observed data set represents in large part the paleosecular variation pattern, it may not necessarily follow a Fisherian distribution.

The deviations from Fisher distribution can cause the angular dispersion of the PG data population to be underestimated. This could explain the low angular dispersion in comparison to the 110-195 Ma curve of Figure 6c.

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Table 1. Paleomagnetic Data for the Ponta Grossa Dikes

Site	Coordinates: Longitude, °W	Coordinates: Latitude, °S	Strike	Width, m	Mean Magnetization Direction: *N/n*	Decl., deg	Incl., deg	α_{95} deg	*k*	Pol	Virtual Geomagnetic Pole: Longitude, deg	Latitude, deg	dm, deg	dp, deg
					Fartura									
1	49.49	23.40	N41°W	30	9/9	352.3	-35.8	2.8	341	N	245.5	82.0	3.2	1.9
2	49.50	23.40	N-S	?	8/8	166.2	30.3	4.3	170	R	66.6	-75.2	4.7	2.6
3	49.50	23.40	N36°W	?	6/6	167.7	40.1	6.2	116	R	41.0	-78.7	7.5	4.5
4	49.50	23.40	N24°W	?	9/9	346.4	-31.8	5.0	108	N	243.9	75.8	5.6	3.1
5[a]	49.50	23.40	?	?	14/14	188.2	-1.5	2.6	234	I	149.9	64.6	2.6	1.3
6	49.50	23.40	?	40	9/9	23.3	-63.3	6.0	73	N	94.7	61.4	9.5	7.5
7	49.50	23.30	N4°E	50	11/8	175.5	51.3	4.0	197	R	334.2	-80.5	5.4	3.6
8	49.50	23.30	?	?	12/12	348.9	-29.8	2.4	331	N	253.4	77.2	2.6	1.5
9	49.40	23.60	N34°E	?	10/10	156.6	62.3	2.2	488	R	348.7	-62.3	3.4	2.7
10	49.50	23.70	?	30	18/18	15.9	-46.5	4.4	63	N	59.8	75.2	5.6	3.6
11	49.70	23.60	?	?	9/6	346.4	-34.7	1.7	1472	N	237.4	76.6	2.0	1.2
12	49.80	23.60	?	50	22/19	165.9	32.6	3.3	105	R	61.7	-75.6	3.7	2.1
13	49.90	23.60	?	100	10/10	164.3	31.4	2.1	550	R	61.4	-73.9	2.3	1.3
14	49.90	23.50	?	20	6/6	344.0	-37.3	1.5	1967	N	227.0	75.0	1.8	1.0
15	49.80	23.50	?	?	4/4	342.6	-34.1	2.0	2032	N	233.3	73.1	2.3	1.3
16[a]	49.60	23.40	N61°W	?	10/10	167.3	-1.5	2.4	413	I	101.6	62.9	2.4	1.2
17[a]	49.60	23.40	?	?	6/6	170.5	-1.2	0.9	5649	I	108.1	64.3	0.9	0.4
18[a]	49.60	23.40	N87°E	?	9/9	167.4	0.5	2.9	322	I	100.8	-63.8	2.9	1.4
19[a]	49.60	23.40	N54°W	10	10/6	173.4	-4.4	4.2	261	I	115.4	63.6	4.2	2.1
21	49.80	23.50	N86°W	100	18/18	348.8	-29.3	1.8	357	N	254.8	76.9	2.0	1.1
22	49.80	23.50	?	15	6/6	349.0	-40.7	2.4	778	N	219.3	79.9	2.9	1.8
23	49.70	23.30	N76°W	50	6/6	345.2	-42.7	4.9	189	N	211.2	76.4	6.0	3.7
84	49.50	23.40	N76°W	100	9/9	164.9	31.5	3.7	198	R	62.0	-74.5	4.1	2.3
119	49.20	23.70	N51°W	?	12/12	4.2	-46.8	4.5	92	N	90.6	84.3	5.9	3.8
120	49.30	23.60	?	70	9/9	158.8	57.5	4.5	130	R	356.9	-66.8	6.6	4.9
121	49.40	23.70	N60°W	50	15/15	9.4	-43.7	1.6	599	N	54.5	81.3	2.0	1.2
					Sapopema									
24	50.50	24.00	N51°W	?	9/9	349.4	-37.8	2.6	382	N	233.3	79.8	3.1	1.8
25	50.50	24.00	?	?	9/9	344.6	-36.3	4.8	115	N	231.5	75.2	5.6	3.3
26[a]	50.50	24.00	N25°W	30										
27	50.50	24.00	N-S	100	9/6	354.5	-44.7	2.3	856	N	193.7	84.5	2.9	1.8
28	50.50	24.00	N65°E	70	21/21	353.8	-42.7	1.8	331	N	210.7	84.3	2.2	1.3
29	50.60	24.00	N45°W	500	6/6	5.9	-31.4	2.3	847	N	348.4	81.1	2.6	1.4
30	50.60	23.90	?	200	14/14	355.4	-50.6	8.0	25	N	157.4	81.5	10.8	7.3
31	50.60	23.70	?	40	12/12	10.2	-26.5	5.0	76	N	355.9	76.3	5.4	2.9
32	50.60	23.70	N35°W	?	10/10	350.5	-29.1	2.8	304	N	260.1	77.9	3.1	1.7
33	50.60	23.70	N60°W	300	9/9	356.9	-34.3	3.9	174	N	278.0	84.3	4.5	2.6
34	50.60	23.80	N35°W	50	10/10	352.2	-33.4	4.6	110	N	255.1	80.9	5.2	3.0
35	50.60	23.80	N20°W	50	9/9	339.2	-28.0	8.1	41	N	240.0	68.5	8.9	4.9
46	50.10	24.40	N45°E	150	6/6	176.4	58.7	6.0	126	R	320.4	-74.7	8.9	6.6
47	50.00	24.70	N45°E	50	9/9	355.4	-52.8	3.3	247	N	153.6	80.4	4.5	3.1
52	50.40	24.50	N30°W	?	11/11	347.8	-42.2	3.1	213	N	217.8	78.9	3.9	2.4
					Telêmaco Borba									
36	51.10	24.10	N51°W	20	16/11	354.5	-35.0	1.5	873	N	260.9	83.0	1.8	1.0
37	51.10	24.10	N80°E	100	9/9	181.2	51.0	4.7	121	R	301.2	-82.3	6.4	4.3
38	51.00	24.00	N20°W	100	6/6	334.8	-40.5	10.3	43	N	216.0	66.9	12.5	7.5
39	51.00	24.10	N75°E	30	9/6	154.4	56.1	2.0	1150	R	3.3	-64.7	2.8	2.0
40	51.00	24.20	N65°E	?	3/3	3.5	-52.9	4.5	757	N	111.4	80.2	6.2	4.3
41	50.80	24.30	N80°W	?	6/6	3.4	-39.1	3.3	404	N	4.9	86.1	4.0	2.4
42	50.70	24.40	N65°W	50	10/10	2.9	-33.8	4.5	118	N	334.7	83.5	5.1	2.9
43	50.70	24.40	?	150	9/9	353.0	-43.1	2.7	372	N	211.7	83.6	3.3	2.1
44	50.60	24.30	N53°W	30	10/10	341.8	-21.0	3.1	246	N	254.0	68.1	3.2	1.7
45	50.40	24.50	N45°W	?	7/7	160.1	57.3	2.4	620	R	356.4	-68.4	3.5	2.6
49	50.40	24.60	N55°W	20	9/9	156.1	57.5	5.2	99	R	0.1	-65.6	7.6	5.6
50	50.50	24.80	N85°W	?	6/6	10.1	-40.6	2.7	598	N	31.7	80.6	3.3	2.0
51	50.50	24.80	N60°W	100	8/8	0.5	-41.2	3.3	286	N	330.9	88.8	4.0	2.4
53	50.90	24.00	N50°W	100	9/9	357.7	-29.0	2.6	402	N	294.6	81.3	2.8	1.6
54	50.90	24.20	?	50	9/9	331.1	-57.5	2.2	568	N	182.7	61.8	3.2	2.3
55	51.00	24.20	?	20	11/11	4.3	-48.2	6.4	52	N	92.0	83.7	8.3	5.5
56	50.80	24.70	?	50	8/6	347.3	-46.6	2.5	701	N	200.9	78.2	3.3	2.1
57	50.80	24.60	N50°E	10	6/6	341.6	-58.0	3.8	311	N	172.7	69.0	5.6	4.1
58	51.30	24.60	N84°E	30	7/7	147.9	46.7	9.3	43	R	24.9	-61.1	11.9	7.7
59	51.30	24.60	?	40	9/9	192.7	21.7	1.3	1460	R	173.0	-72.0	1.4	0.8

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Table 1. (continued)

Site	Coordinates		Strike	Width, m	Mean Magnetization Direction					Pol	Virtual Geomagnetic Pole			
	Longitude, °W	Latitude, °S			N/n	Decl., deg	Incl., deg	α_{95} deg	k		Longitude, deg	Latitude, deg	dm, deg	dp, deg
					Telêmaco Borba (continued)									
60	51.50	24.50	N41°E	100	9/9	359.5	-38.2	1.5	1123	N	299.8	86.9	1.8	1.1
61	51.80	24.50	N34°W	50	9/9	359.7	-56.7	1.8	807	N	129.3	77.2	2.6	1.9
62	50.80	24.50	?	?	56/43	186.8	39.5	1.7	167	R	202.3	-83.4	2.0	1.2
63	50.70	24.50	?	20	3/3	2.0	-54.2	2.2	3221	N	120.2	79.7	3.1	2.1
64	50.70	24.50	N85°E	?	8/8	358.0	-50.7	5.6	100	N	143.0	82.9	7.5	5.0
65	50.70	24.60	?	50	9/9	161.9	45.4	1.8	826	R	27.5	-73.6	2.3	1.4
66	50.70	24.60	N40°E	20	9/9	357.5	-35.0	4.0	166	N	285.1	84.2	4.6	2.7
67	50.60	24.60	N80°W	20	9/9	335.3	-37.4	8.3	39	N	223.6	67.0	9.8	5.7
68	50.60	24.70	N25°W	20	9/9	358.5	-40.6	1.0	2719	N	266.4	87.9	1.2	0.7
69	50.60	24.70	N75°E	100	9/9	4.2	-33.4	2.4	453	N	341.6	82.5	2.8	1.6
70	50.60	24.70	N75°E	30	9/9	357.6	-43.4	1.5	1151	N	203.8	87.7	1.9	1.2
71	50.60	24.70	N75°E	30	12/12	353.3	-36.6	2.1	439	N	253.3	82.5	2.4	1.4
					Curitiba									
72	49.90	25.00	N40°W	0.3	6/3	335.8	-28.1	4.4	780	N	239.4	65.2	4.8	2.7
73[a]	49.90	25.10	?	?										
74	49.90	25.10	?	?	12/12	345.1	-33.3	8.0	30	N	243.8	74.6	9.1	5.2
75	49.90	25.10	N60°W	?	9/9	0.5	-51.4	2.2	533	N	126.7	83.0	3.0	2.1
76	49.90	25.10	N-S	5	6/6	0.3	-35.2	3.2	430	N	313.3	84.3	3.7	2.2
77	49.80	25.20	N40°W	30	9/9	160.6	54.2	2.2	561	R	6.1	-70.7	3.1	2.1
78	49.80	25.20	?	20	6/6	341.8	-42.0	5.7	141	N	219.6	73.5	6.9	4.3
79	48.90	25.60	N46°W	100	9/9	154.0	37.6	4.7	123	R	46.4	-65.8	5.5	3.2
79A	48.90	25.60	N71°W	80	6/6	351.0	-38.2	2.2	930	N	245.7	80.7	2.6	1.5
80	48.80	25.60	E-W	20	9/9	357.6	-42.8	2.6	405	N	239.3	87.7	3.2	2.0
81	48.80	25.60	?	?	6/6	165.0	46.3	2.7	617	R	29.2	-76.4	3.5	2.2
82	48.80	25.50	N70°W	10	12/12	349.8	-32.1	2.7	253	N	259.7	77.6	3.1	1.7
83	48.80	25.30	?	?	12/12	181.2	45.2	1.7	691	R	273.6	-88.2	2.1	1.3
85	48.90	25.20	N76°W	100	16/16	354.4	-58.0	3.9	90	N	149.1	75.7	5.7	4.2
86	48.90	25.20	N70°W	80	6/6	171.2	45.0	4.8	197	R	29.6	-81.9	6.1	3.8
87	49.10	25.10	N56°W	30	5/5	181.1	46.2	4.5	293	R	289.4	-87.4	5.7	3.7
88[a]	49.00	24.70	?	10										
89[a]	49.30	24.80	?	20										
90	49.30	25.00	?	?	5/5	174.4	44.2	2.4	1017	R	29.4	-84.9	3.0	1.9
91	49.30	25.10	?	40	12/12	168.5	44.5	1.6	723	R	32.1	-79.6	2.0	1.3
92	49.40	25.10	N30°W	40	3/3	180.4	37.3	1.9	4454	R	135.9	-85.8	2.2	1.3
93	49.40	25.10	N44°E	20	6/6	346.4	-34.0	3.9	301	N	245.0	75.8	4.4	2.5
94	49.30	25.20	?	?	6/6	164.7	48.9	2.7	633	R	18.6	-75.7	3.5	2.3
95	49.30	25.20	N74°E	120	9/9	161.6	33.4	2.3	509	R	59.1	-71.6	2.6	1.5
96	49.20	25.30	N66°W	?	12/12	353.0	-54.6	2.1	444	N	160.9	78.4	2.9	2.1
97	49.20	25.30	N66°W	?	4/4	351.4	-53.1	7.2	163	N	170.6	78.8	10.0	6.9
98	49.20	25.30	N56W	?	5/5	192.3	41.6	4.7	270	R	216.5	-78.8	5.7	3.5
99	49.30	25.30	N30W	40	9/9	344.3	-33.7	1.6	1084	N	242.9	73.9	1.8	1.0
100	49.30	25.40	N30W	20	7/7	11.1	-54.7	7.1	72	N	89.3	76.3	10.1	7.1
101	49.30	25.40	N60W	60	11/11	343.0	-49.4	3.0	229	N	199.0	74.3	4.0	2.7
102	49.20	25.30	N41W	70	6/6	349.6	-51.1	1.5	2098	N	183.0	78.8	2.0	1.3
103	49.20	25.20	N71W	60	4/4	154.6	28.5	3.8	581	R	58.8	-64.2	4.2	2.3
104	49.00	24.60	?	150	27/27	358.4	-46.0	1.9	218	N	158.7	86.9	2.4	1.5
122	49.10	25.80	N30°E	4	9/9	355.8	-54.1	1.4	1404	N	152.4	80.4	1.9	1.4
123	49.10	25.80	N30°E	10	6/6	348.4	-53.1	2.1	1066	N	180.3	77.2	2.8	2.0
					Guapiara									
105	48.50	25.10	N67°W	50	12/12	165.8	34.8	1.8	564	R	62.9	-75.6	2.1	1.2
106	48.60	24.20	N60°W	?	8/4	336.6	-39.8	8.4	120	N	220.8	68.5	10.1	6.1
107	48.60	24.20	N32°W	200	9/6	345.1	-49.8	1.6	1802	N	192.4	75.3	2.1	1.4
108	48.60	24.20	?	?	3/3	352.4	-32.8	1.4	7810	N	261.7	80.4	1.6	0.9
109	48.60	24.20	?	100	9/9	354.9	-34.8	1.3	1522	N	267.3	83.1	1.5	0.9
110	48.70	24.20	?	?	3/3	1.0	-39.0	15.1	68	N	334.7	87.6	18.0	10.7
111	48.70	24.20	?	?	10/6	1.7	-24.9	11.4	35	N	320.0	78.8	12.3	6.6
112	48.70	24.20	?	?	7/7	358.3	-28.0	4.6	174	N	301.5	80.6	5.0	2.8
113	48.80	24.20	?	30	9/9	171.8	52.8	3.5	219	R	347.3	-78.3	4.8	3.3
114	48.80	24.30	N47°W	100	12/12	196.2	32.3	6.9	40	R	200.3	-73.4	7.8	4.4
115[a]	48.80	24.30	?	3										
116[a]	48.90	24.50	?	?										

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Table 1. (continued)

Site	Coordinates		Strike	Width, m	Mean Magnetization Direction					Pol	Virtual Geomagnetic Pole			
	Longitude, °W	Latitude, °S			*N/n*	Decl., deg	Incl., deg	α_{95} deg	*k*		Longitude, deg	Latitude, deg	dm, deg	dp, deg
Guapiara (continued)														
117	48.80	24.20	N35°E	?	4/4	340.0	-49.9	3.1	878	N	196.7	71.1	4.1	2.8
118[a]	48.80	24.20	?	10	12/8	174.1	-4.1	15.9	13	I	118.1	63.1	16.0	8.0
131	49.60	24.00	N35°W	?	9/9	164.0	55.7	2.3	516	R	355.0	-71.5	3.3	2.3
132	48.10	24.60	N78°E	?	9/9	165.0	51.0	2.1	612	R	10.3	-75.0	2.8	1.9
133	49.50	24.60	N7°W	?	6/6	352.6	-49.2	0.8	7297	N	179.0	81.4	1.0	0.7
134	48.90	24.60	N63°E	?	3/3	345.3	-43.0	5.4	519	N	216.4	76.7	6.7	4.2
135	49.10	24.80	?	?	8/5	144.9	51.3	3.6	457	R	19.6	-58.4	4.9	3.3

N is number of specimens per site; *n* is number of specimens included in the mean; Decl and Incl is declination and inclination; α_{95} and *k* are *Fisher's* [1953] statistical parameters; dp and dm define the 95% confidence oval for the virtual geomagnetic poles; and Pol is polarization (N is normal, R is reversed, and I is intermediate). Blank entries indicate no consistent results.
*Not included in determination of paleomagnetic pole.

Butler et al. [1991] suggest that if the reversal test is positive, then the geomagnetic secular variation during both normal-polarity and reversed-polarity intervals has been adequately sampled by a data set. The reversal test has been performed here following the *McFadden and Lowes* [1981] method. The mean of the normal polarity directions is not distinguishable from the mean of the reversed polarity directions at the 95% confidence level, indicating that the reversal test is positive.

From the preceding discussions it is concluded that the

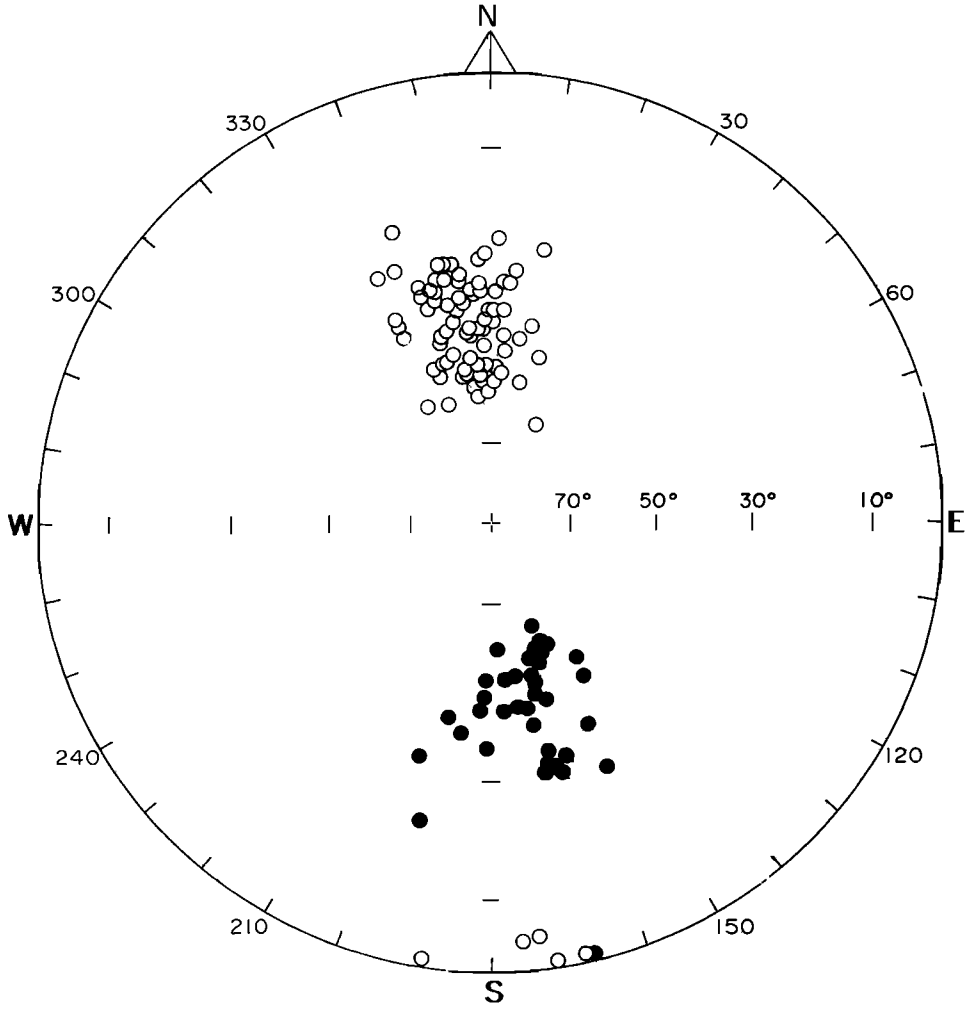


Figure 4. Mean magnetization directions for 121 PG dikes plotted on Wulff stereogram. Open and solid symbols correpond to negative and positive inclinations, respectively.

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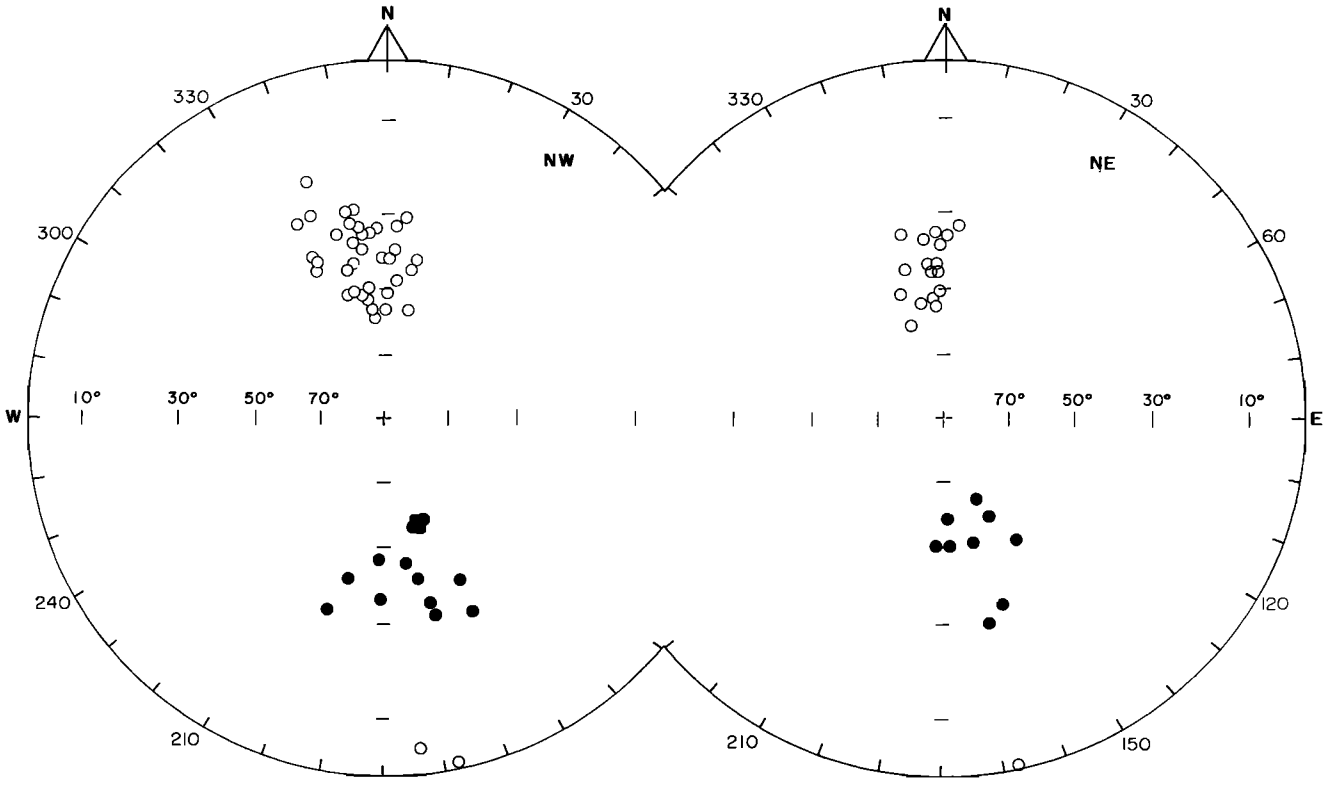


Figure 5. Mean magnetization directions of the NW and NE trending dikes plotted on Wulff stereogram. Open and solid symbols are as in Figure 4.

Table 2. Selected Mesozoic Paleomagnetic Poles for South America

Formation	Code	Age, Ma	Paleomagnetic Pole					Reference
			N	Longitude, °E	Latitude, °S	α_{95} deg	k	
Patagonia, Argentina-Chile	PT	79-64	18	358.4	78.7	6.5	32	*Butler et al.* [1991]
Volcanics, central Argentina	VC	85-66	12	45.0	70.0	12.1	14	*Valencio et al.* [1983]
Passa Quatro-Itatiaia	PQ	73-70	18	360.0	80.0	6.0	34	*Montes-Lauar* [1993]
Poços de Caldas	PC	83	47	320.1	83.2	2.8	55	*Montes-Lauar* [1988]
Cabo de Santo Agostinho	SA	99-85	9	315.0	87.6	4.5	127	*Schult and Guerreiro* [1980]
Pirgua Subgroup	PI	114-77	5	306.7	89.4	17.8	19	*Valencio et al.* [1977]
Rio Los Molinos dikes	LM	150-63	7	13.0	78.0	8.5	51	*Linares and Valencio* [1975]
Sierra de Los Condores[a]	LC	129-119	27	31.0	79.1	4.7	35	*Mendia* [1978], *Vilas* [1976], *Valencio* [1972]
Maranhão intrusions	MI	130-120	21	81.0	83.6	1.9	276	*Schult and Guerreiro* [1979]
Central alkaline province	CAP	130-120	75	62.3	85.4	3.1	29	*Ernesto et al.* [1995]
Ponta Grossa dikes	PG	135-127[b]	115	30.3	82.4	2.0	44	*this paper*
Southern Paraná Basin[a]	SPB	133±1[c]	197	106.2	84.0	1.5	45	*Ernesto et al.* [1990]
Central Paraná Basin[a]	CPB	133-132[c]	106	64.4	84.1	2.3	38	*Ernesto et al.* [1990]
Northern Paraná Basin[a]	NPB	137-132[b,c]	36	77.1	83.2	4.5	29	*Ernesto et al.* [1990]
Rio Negro Formation	RN	Neoc/Bar	8	135.6	80.1	—	—	*Castillo et al.* [1991]
Chon-Aike Formation	CA	172-167	54	197.0	85.0	6.0	10	*Vilas* [1974]
Maranhão Flows	MF	175±2[e]	15	262.7	85.3	7.0	32	*Schult and Guerreiro* [1979]
Marifil Formation	MR	Jm	14	145.4	83.1	15.0	—	*Rapalini et al.* [1993]

N is number of sites; α_{95} and k are *Fisher's* [1953] statistical parameters.
[a] Recalculated in this paper.
[b] $^{40}Ar/^{39}Ar$ ages [*Turner et al.*, 1994].
[c] $^{40}Ar/^{39}Ar$ ages [*Renne et al.*, 1992a].
[d] $^{40}Ar/^{39}Ar$ ages [*Renne et al.*, 1992b].
[e] $^{40}Ar/^{39}Ar$ ages [*Laurenzi*, 1991].

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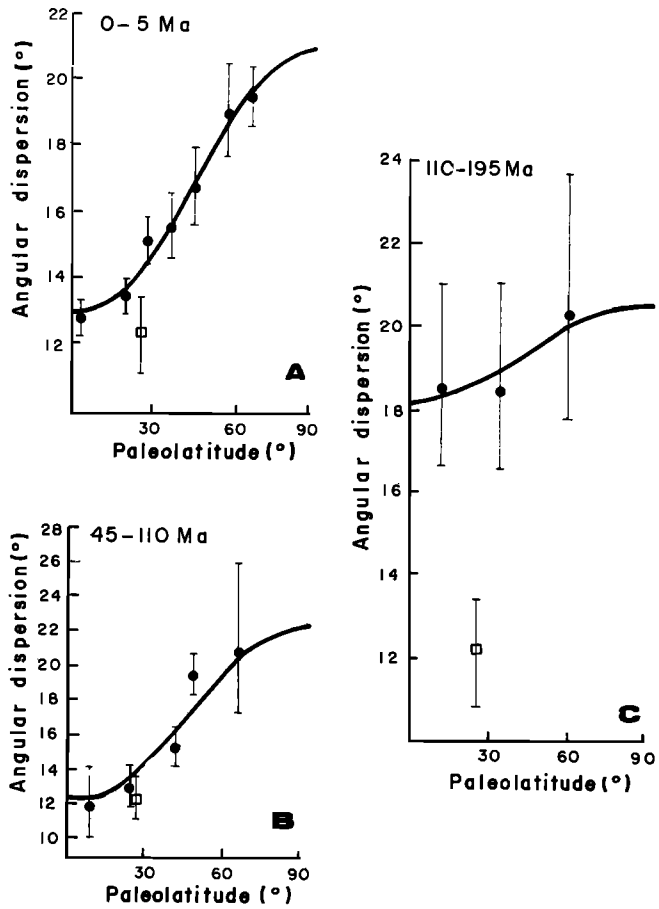


Figure 6. Angular dispersion curves from *McFadden and McElhinny* [1984]: (a) 0-5 Ma, (b) 45-110 Ma, and (c) 110-195 Ma. The open square represents the angular dispersion calculated for PG dikes. Published with permission from Blackwell Science.

geomagnetic secular variations were adequately sampled by the 115 studied dikes and their effects were conveniently averaged from the VGP mean, allowing the calculation of a very well defined Lower Cretaceous reference pole.

Discussion

For comparison the PG pole is plotted in Figure 9 along with the Paraná Basin (Serra Geral Formation) paleomagnetic poles presented by *Ernesto et al.* [1990, Table 22]. The SD pole presented in that paper is based on data from four Ponta Grossa dikes and six sills intruding prevolcanic sediments in the northeastern border of the Paraná Basin. This pole has been recalculated here (40.1°E, 77.1°S; $N = 6$, $\alpha_{95} = 13.5°$, $k = 25$) considering only data from the sills since those Ponta Grossa dikes have been included in the new PG pole (this paper) after reanalyzing the samples following the same methodology adopted in this work (sites 131 to 137 in Table 1). In Figure 9, the Paraná Basin poles are discriminated according to the three different regions in the basin, as subdivided by *Bellieni et al.* [1984] based on the distribution of the chemically varying rock types. In this way, we will hereinafter refer to the southern (SPB, south of the Rio Uruguai lineament (Figure 1)), northern (NPB, north of Rio Piquiri lineament (Figure 1)), and central (CPB, comprised between SPB and NPB) Paraná Basin. The NPB corresponds to the major area of the basin and includes the Ponta Grossa Arch and the aforementioned SD sills.

In Figure 9 the PG pole is relatively far from the clustering of the SPB poles but lies close to some CPB and NPB poles.The large α_{95} confidence envelope seen in the figure is determined by the confidence circles of NPB poles. However, some considerations about the reliability of individual Paraná Basin poles must be addressed. The SPB poles are tightly grouped, while the CPB and NPB poles are far more scattered. The reason for this behavior is the number of sites used to compute the mean of VGPs. All SPB poles are based on at least 10 time independent sites with very good stratigraphic control, as they were sampled in the escarpments of the Serra Geral Formation. In the other two regions, sampling conditions were not so favorable, especially in NPB, where all poles are based on fewer than 10 sites. This is an indication that the minimum of three or five independent determinations of the ancient geomagnetic field, accepted as a reliability criterion [e.g., *Beck*, 1988], may actually be insufficient in the case of these volcanic rocks. Because of the intermittent character of the magmatic activity, the difficulties in averaging out the secular variation effects may be greater in igneous than in sedimentary rocks. Three or even more sites may represent a rapid succession of magmatic "pulses" in a very brief time interval, and rocks will actually record the same secular variation phase. *Ernesto* [1977] had already noticed that not all site mean directions in some Serra Geral flow sequences represented time-independent data. As a result, the angular dispersion calculated for those data set was lower than the expected angular dispersion predicted by the geomagnetic secular variation models. However, after properly filtering the data by means of the *Watson and Beran* [1967] method the data set showed the expected angular dispersion indicating that those flow sequences (those with more than 10 sites) conveniently average the secular variation. *Calderone et al.* [1990], investigating volcanic rocks in the Mojave-Sonora Desert region, also reported the influence of rapid extrusion on the averaging of secular variation.

We therefore conclude that the Paraná Basin paleomagnetic poles based on fewer than 10 sites may not have adequately averaged secular variation. However, the mean poles for the three regions in the basin computed giving unit weight to individual poles tend to minimize this effect. For the purposes of this paper we are recalculating those poles, assigning unit weight to sampling sites or flows in order to make the associated statistical parameters compatible to those obtained with the large Ponta Grossa data set. The new computed poles are located at SPB (106.2°E, 84.0°S; $N = 197$; $\alpha_{95} = 1.5°$; $k = 45$), CPB (64.6°E, 84.1°S; $N = 106$; $\alpha_{95} = 2.3°$; $k = 38$) and NPB (77.1°E, 83.2°S; $N = 36$; $\alpha_{95} = 4.5°$; $k = 29$) and are plotted in Figure 10. These poles are not statistically distinguishable at the 95% confidence level; however it must be noted that NPB is still poorly defined, based on few sites; its statistical parameters demonstrate that the data are not conveniently distributed.

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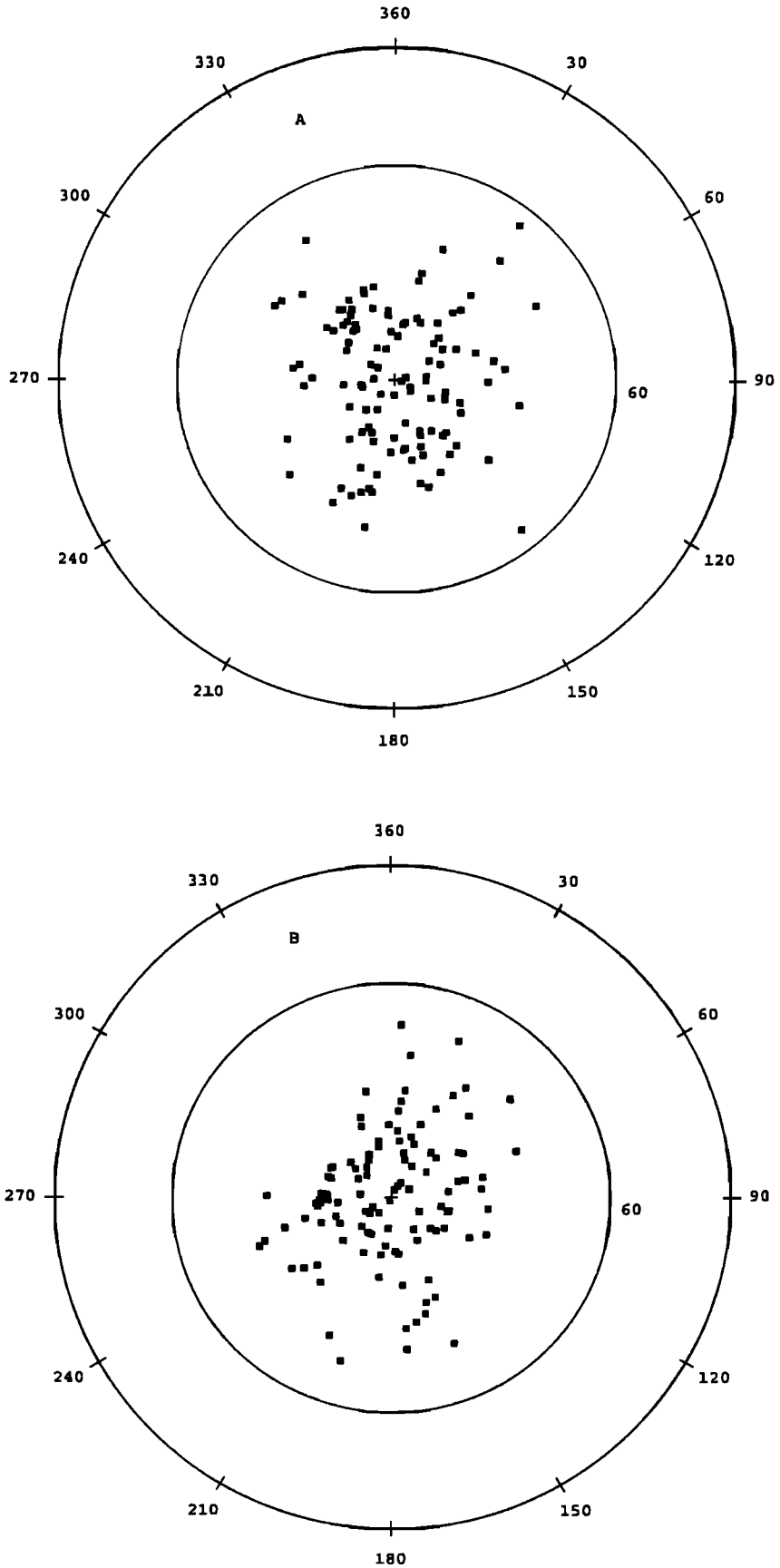


Figure 7. Centered stereographic plots for 115 PG dikes. (a) Magnetization directions normalized for inclination < 0, (b) VGPs in the southern hemisphere.

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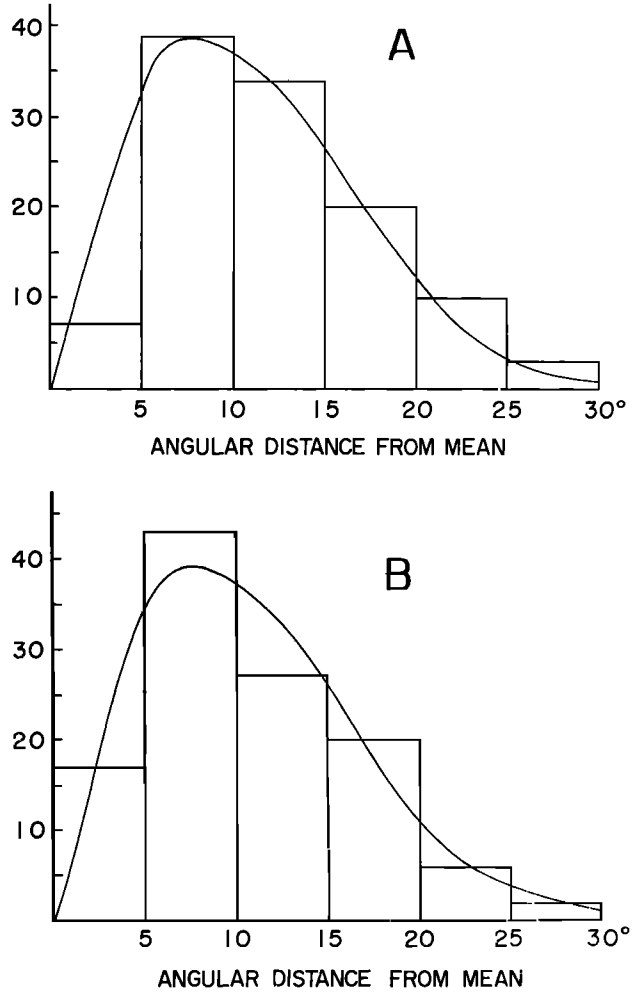


Figure 8. Comparison of the observed angular frequency distribution (histogram) and the expected Fisherian distribution (curve). (a) Magnetization directions, (b) VGPs.

Detailed $^{40}Ar/^{39}Ar$ determinations [*Renne et al.*, 1992a], including data from the bottom and top of the flow sequences, give a SPB mean age of 133±1 Ma, whereas ages from the CPB range from 132.0±0.4 to 132.9±0.6 Ma [*Renne et al.*, 1992b], tending to be slightly younger than SPB. For the NPB, *Turner et al.* [1994] report ages varying from 137±0.7 to 128.7±1.1 Ma obtained mainly from borehole samples; however, the oldest reported ages correspond to basal samples and may not represent the age of the surface samples used in the paleomagnetic work. On the other hand, it is possible that the scattering of the NPB paleomagnetic data reflects this longer age interval.

As can be seen in Figure 10, the confidence circle of PG pole does not overlap the other Paraná Basin poles (except for a minor superposition on the NPB circle); the PG pole is distinguishable from the Paraná Basin poles at the 95% confidence level [*Raposo*, 1992]. These mean poles are distributed along a small circle (84°S) and seem to delineate along with the PG pole a segment of the apparent polar wander path for the Early Cretaceous. However, considering the apparentlly small time interval (see Table 2) they comprise, this polar shift may imply in a very fast continental drift.

Considering a unique paleomagnetic pole for the three chemically defined regions in the Paraná Basin (Serra Geral Formation pole in Figure 10; SG pole at 90.1°E, 84.3°S; $N = 339$, $\alpha_{95} = 1.2°$, $k = 40$) the resulting plate motion since SG times until the emplacement of the PG dikes would be a clockwise rotation of about 4.5° within error limits of $\Delta = \pm 3.2$ [*McWilliams*, 1984] and flattening of -7.5±2.3. This rotation is consistent with the predicted movements of the South American plate at the initial stages of the South Atlantic opening. The rift evolved from south to north, reaching latitudes of about 38°S at 130 Ma [*Nürnberg and Müller*, 1991] and causing a "scissorslike" opening between South America and Africa, with South America rotating clockwise. The total displacement prescribed by the arch between SG and PG is 6.9°. If we consider the 10 m.yr. time interval for the duration of the whole magmatism in the Paraná Basin (including the Ponta Grossa dikes) proposed by *Turner et al.* [1994] the required velocity would be of approximately 0.7°/m.yr. (~7 cm/yr). Although this rate is compatible with present velocities of 6-10 cm/yr. for plates with a significant amount of continental crust [*Meert et al.*, 1993], the higher spreading half rate reported for the South Atlantic is of the order of 3.8 cm/yr [*Nürnberg and Müller*, 1991] between Chron M4 (126.5 Ma) and Chron 34 (84 Ma). However, anomaly M4 is the oldest clearly identifiable magnetic anomaly on oceanic crust of the South American and African plates, and therefore plate velocities prior to the effective creation of oceanic crust are difficult to ascertain. It must be taken into account that a significant amount of the APWP indicated by our data set may be due to true polar wander (TPW), as *Besse and Courtillot* [1991] reported TPW over 20° in the last 200 Ma, although these results are still speculative because a fixed hotspot reference frame is assumed.

It is commonly proposed that intracontinental deformations occurred in the South American and African plates as a consequence of the evolution of the South Atlantic Ocean. These deformations could have affected the paleomagnetic data. *Curie* [1984] proposed a right-lateral transcontinental strike-slip movement across the Paraná Basin in continuation to the Walvis-São Paulo ridges, and *Conceição et al.* [1988] proposed that this discontinuity ran along the NW trending Curitiba-Maringá shear zone along the Ponta Grossa Arch. However, these deformations predate the emplacement of the Ponta Grossa dikes, and no other field evidence of important tectonism occurring during or after the emplacement of these dikes is observed. In fact, *Chang et al.* [1992] argue that those assumptions lack of field evidences and suggest a model involving strain accommodation by rotation of South America relative to Africa, producing extension in a NE-SW direction, and injection of NW-SE oriented dikes. Furthermore, *Rapalini et al.* [1993] found no effects of intracontinental movements within the South American plate affecting paleomagnetic data from Late Carboniferous to Middle-Late Jurassic, by comparing statistically the data from northern and southern South America.

The PG pole and the three recalculated poles for the Paraná Basin (SPB, CPB, and NPB) are plotted in Figure 11 along with other Mesozoic paleomagnetic poles for stable South America (Table 2). Mean poles for Middle-

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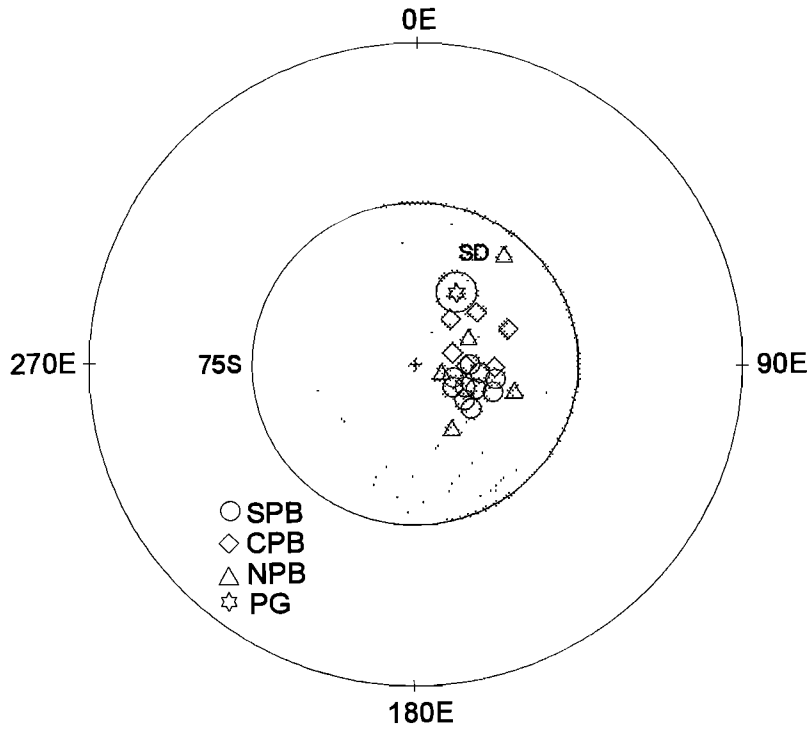


Figure 9. Comparison between the paleomagnetic poles for PG dikes and Paraná Basin volcanic rocks. SPB, CPB, and NPB correspond to the chemically defined southern, central and northern Paraná regions [*Bellieni et al.*, 1984], respectively; SD is the recalculated pole for the sills from NPB. Shadowed area indicates the α_{95} envelope.

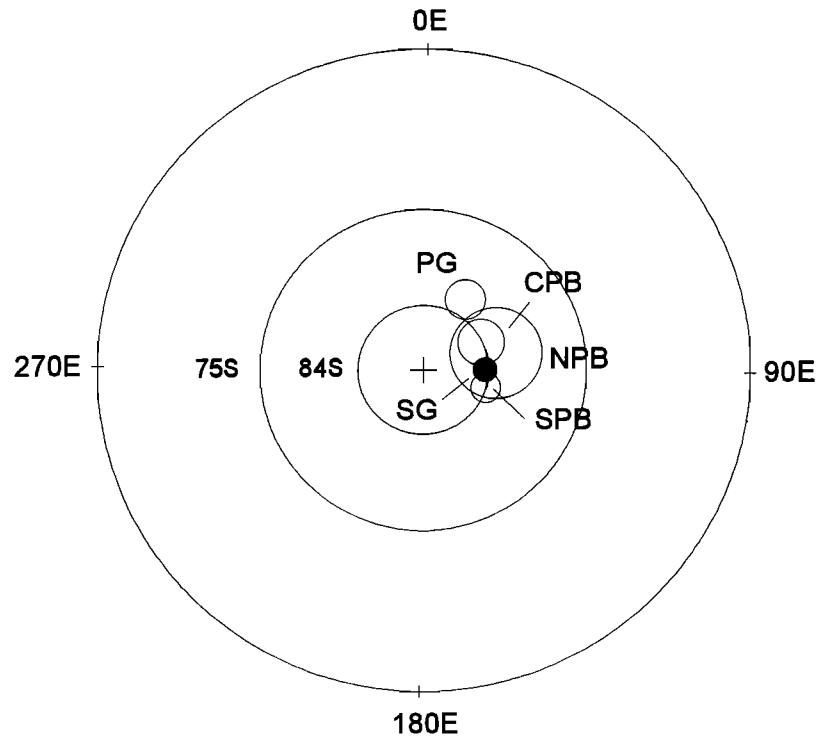


Figure 10. Paleomagnetic poles for the southern (SPB), central (CPB) and northern (NPB) Paraná Basin regions (Table 2) and the Ponta Grossa dikes (PG) represented by their α_{95} confidence circles (open circles). The solid circle is the α_{95} confidence parameter for the SG pole calculated from all sites of the Paraná Basin regions.

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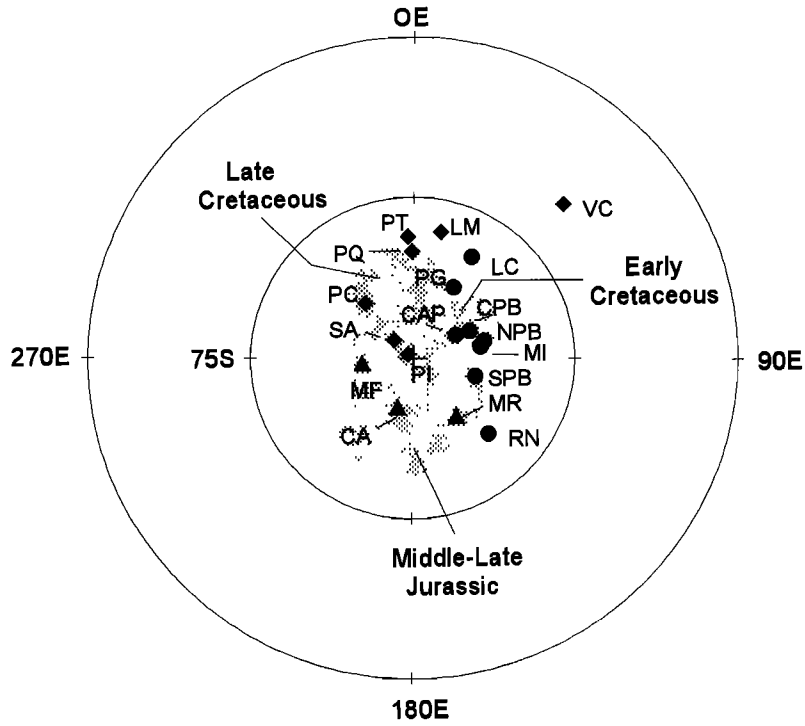


Figure 11. Middle-Late Jurassic (triangles), Early Cretaceous (circles) and Late Cretaceous (diamonds) selected paleomagnetic poles for South America (Table 2) and corresponding mean poles with their α_{95} confidence circles.

Late Jurassic (191.3°E, 86.2°S; $N = 3$, $\alpha_{95} = 7.8°$), Early Cretaceous (60.3°E, 83.8°S; $N = 7$, $\alpha_{95} = 2.9°$; including PG pole) and Late Cretaceous (346.5°E, 84.1°S; $N = 5$, $\alpha_{95} = 4.9°$) were computed and represented in Figure 11 by their confidence circles. For the calculation of the Late Cretaceous mean, pole VC (Table 2) has been excluded because its position seems unreliable compared with the other poles, which is reinforced by its poor statistical parameters; pole LM was also discarded as its age is highly uncertain. For the Early Cretaceous mean pole we preferred to use SPB, CPB, and NPB individually instead of the SG pole in order to minimize the effect of less reliable poles; RN was discarded because no statistical parameters are reported. Although data for Middle-Late Jurassic are still scarce, it is clearly seen in Figure 11 that the three mean Mesozoic poles are well discriminated and that the adoption of a unique mean pole comprising data from Middle Jurassic to Late Cretaceous is not justifiable. During this time interval, South America was undergoing a more complex movement than a simply east-west drift.

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